

GRADED PROJECT-1

Graded Lab: DNS Analysis and Troubleshooting Using Wireshark in Kali Linux

Objective:

In this lab, you will simulate a real-time DNS issue on a network and use **Wireshark** on **Kali Linux** to capture, analyze, and troubleshoot DNS traffic. You are expected to identify the root cause of the DNS issue and provide a solution.

Scenario:

You are a network administrator responsible for managing a company's internal DNS `server. Several users have reported slow website access and intermittent failures when resolving domain names. Your task is to troubleshoot the DNS problem using Wireshark and Kali Linux, identify the root cause, and recommend a fix.

Lab Setup:

1. **Kali Linux:** This will be your analysis machine, running Wireshark for packet capture and analysis.
 2. **DNS Server:** A server on the network responsible for resolving domain names.
 3. **Client Machines:** Users are experiencing intermittent issues with DNS resolution.
 4. **Internet Connection:** Required to simulate real-world DNS queries.
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Grading Criteria:

- **DNS Packet Capture and Analysis (30%):** Ability to correctly capture DNS traffic and analyze it using Wireshark.
- **Identifying the Issue (30%):** Ability to identify the root cause of DNS resolution failure.
- **Solution Recommendation (20%):** Ability to suggest a valid fix to resolve the DNS issue.
- **Report Submission (20%):** Clear and concise lab report, explaining steps taken, findings, and proposed solutions.